FINS3645: Financial Market Data Design & Analysis

Part B Report: Funds, Sentiment & App

Stations 3 & 4: Model Design and Implementation

App: "Quantise"

App: https://z5488988-projectb.streamlit.app

Repo: https://github.com/cynd-anhtran/z5488988_projectB

zID: z5488988

Name: Anh Thu Tran

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1. The Funds and the Backtest Design

1.1 Fund Families

Quantise presents twelve base funds across three asset families. The Equity family invests in 50 US large-cap stocks from the 10 sector groups supplied with the data. The Crypto family covers 10 major cryptocurrencies (BTC-USD, ETH-USD, BNB-USD, XRP-USD, ADA-USD, DOGE-USD, SOL-USD, DOT-USD, LTC-USD, and BCH-USD), while Combined includes all 60 assets. The analysis uses the Equity price, Crypto price, and news datasets supplied in the FINS3645 project bundle, with the cleaned headline panel carried forward from Part A. Returns are first calculated on each market's own calendar. For Combined, the already-calculated Crypto returns are selected on Equity trading dates; weekend-only returns remain in the Crypto fund but not the Combined fund. Calculating returns before this alignment prevents weekend Crypto movements from being compressed into one Equity trading-day return.

1.2 Optimisation Methods

Four portfolio construction methods are implemented, all subject to long-only, fully invested constraints (weights ≥ 0, summing to 1):

Equal-weight (1/N) assigns the same target weight to every asset at each monthly rebalance. It is the transparent benchmark used by DeMiguel, Garlappi, and Uppal (2009). The method needs no estimates of expected returns or covariances, although its realised performance still depends on the sample and on weight drift between rebalances.

Minimum-variance chooses long-only weights that minimise estimated portfolio variance, $w'\Sigma w$. The implementation uses SLSQP with the analytic gradient $2\Sigma w$. Because it does not target a particular expected return, it avoids direct reliance on mean-return estimates, which are difficult to estimate precisely (Markowitz, 1952; Merton, 1980).

Maximum-Sharpe (tangency) seeks the highest estimated excess return per unit of volatility. The code solves an equivalent transformed problem: minimise $y'\Sigma y$ subject to $(\mu - r_f)'y = 1$ and $y \geq 0$, then normalise y into fully invested weights. If no asset has a positive estimated excess return, the method uses equal weights rather than returning an infeasible solution.

Risk parity aims to give each asset an equal contribution to total portfolio variance (Maillard, Roncalli, & Teïletche, 2010). The code minimises $0.5w'\Sigma w - (1/n)\Sigma\log(w_i)$ with L-BFGS-B and rescales the positive solution to sum to one. It uses covariance estimates but not expected returns.

1.3 Walk-Forward Backtest Design

All funds are evaluated out of sample with a monthly, expanding-window backtest. At the start of each live month, the optimiser uses only observations from earlier dates and sets target weights for that month. Positions are then held without another rebalance, so their weights drift with daily asset returns until the next scheduled rebalance. The return earned in a month is never used to set that month's weights.

The initial estimation window matches the data calendar: 252 trading-day observations for Equity and Combined and 365 daily observations for Crypto. Because Crypto trades seven days a week, its 365 observations are calendar-day observations, not 365 Equity trading days. The realised OOS samples are therefore 4 January 2021 to 29 December 2023 for Equity and Combined (753 observations) and 1 January 2021 to 31 December 2023 for Crypto (1,095 observations).

Performance annualisation follows each fund's sampling frequency: 252 for Equity and Combined and 365 for Crypto. Annualised return is the geometric compound return over the sample, annualised volatility scales the daily standard deviation by $\sqrt{252}$ or $\sqrt{365}$, and Sharpe and Sortino ratios use the same calendar factor. A zero risk-free rate is used for Maximum-Sharpe weights and every reported Sharpe ratio. The standalone Crypto results retain weekend observations and are not evaluated with an Equity convention.

After each optimisation, the code checks solver success and verifies that the weights are finite, non-negative, and sum to one. It also checks the transformed return constraint for Maximum-Sharpe and the equal-risk-contribution condition for Risk parity. Invalid solutions raise an error instead of entering the results silently.

2. Out-of-Sample Results and Fund Fact Sheets

2.1 Performance Overview

Table 1 reports the gross OOS results for all twelve funds. Equity Sharpe ratios range from 0.59 to 0.82, Crypto from 0.54 to 0.84, and Combined from 0.40 to 0.78. Every Crypto method has a positive compound return, but annualised volatility of 65%–82% and maximum drawdowns of 76%–85% show why return alone is not enough to compare these funds.

Table 1. Gross out-of-sample performance across all fund families and methods. Equity and Combined use 252-day annualisation and run from 2021-01-04 to 2023-12-29; Crypto uses 365-day annualisation and runs from 2021-01-01 to 2023-12-31.

Universe	Method	Ann. Ret.	Ann. Vol.	Sharpe	Sortino	Max DD	Total Ret.
Equity	Equal-weight	12.64%	16.12%	0.82	1.18	−20.25%	42.72%
Equity	Min-variance	6.92%	12.75%	0.59	0.84	−17.99%	22.14%
Equity	Max-Sharpe	13.77%	23.66%	0.66	0.99	−37.67%	47.04%
Equity	Risk parity	10.07%	14.93%	0.72	1.03	−19.40%	33.20%
Crypto	Equal-weight	40.50%	81.89%	0.83	1.19	−81.57%	177.38%
Crypto	Min-variance	27.16%	65.19%	0.70	1.02	−75.56%	105.61%
Crypto	Max-Sharpe	12.08%	79.36%	0.54	0.77	−85.07%	40.78%
Crypto	Risk parity	41.79%	80.26%	0.84	1.21	−80.47%	185.06%
Combined	Equal-weight	15.14%	21.60%	0.76	1.08	−27.87%	52.37%
Combined	Min-variance	6.92%	12.75%	0.59	0.84	−17.99%	22.14%
Combined	Max-Sharpe	8.13%	34.90%	0.40	0.58	−52.10%	26.31%
Combined	Risk parity	12.79%	17.44%	0.78	1.11	−22.38%	43.28%

2.2 Equity Fund Interpretation

Within Equity, Equal-weight records the highest Sharpe ratio (0.82) and Sortino ratio (1.18). Maximum-Sharpe has the highest annualised return (13.77%) but also the highest volatility (23.66%) and deepest drawdown (−37.67%). Minimum-variance has the lowest volatility (12.75%) and shallowest drawdown (−17.99%), alongside a lower 6.92% annualised return. The benchmark's strong risk-adjusted result is consistent with evidence that estimation error can erode the realised advantage of optimised portfolios (DeMiguel et al., 2009); it is specific to this sample rather than proof that 1/N always dominates.

2.3 Crypto Fund Interpretation

Crypto Risk parity produces the highest annualised return (41.79%) and Sharpe ratio (0.84), closely followed by Equal-weight (40.50%, Sharpe 0.83). Minimum-variance lowers annualised volatility to 65.19% and records the least severe maximum drawdown (−75.56%), although the remaining downside risk is still large. Maximum-Sharpe performs worst on a risk-adjusted basis (12.08% annualised return, Sharpe 0.54, maximum drawdown −85.07%). In this sample, using estimated mean returns did not improve the Crypto risk–return trade-off, consistent with the difficulty of estimating expected returns from a short and volatile history (Merton, 1980). These results describe 2021–2023 and are not forecasts.

2.4 Combined Fund Interpretation

Combined Equal-weight earns 15.14% annualised with a Sharpe ratio of 0.76, while Combined Risk parity delivers the family's highest Sharpe (0.78) and Sortino ratio (1.11) with a smaller −22.38% maximum drawdown. A wider universe does not improve every method: Combined Maximum-Sharpe records an 8.13% annualised return, 34.90% volatility, and a −52.10% maximum drawdown. Combined Minimum-variance assigns the Crypto sleeve zero weight within numerical precision (the maximum monthly Crypto sleeve is below 1×10^{-17}). Its results are therefore identical to Equity Minimum-variance at the precision shown in Table 1. This is an optimiser outcome, not duplicated data.

2.5 Portfolio Results in Figures

Figures 1–4 report OOS growth, drawdown, risk-adjusted performance, and changing weights. Each figure uses the corrected return calendar and the same precomputed results reported in Table 1. The deployed app also provides a selectable fact sheet for each of the 12 family–method funds, including the latest historical target holdings from the most recent rebalance.

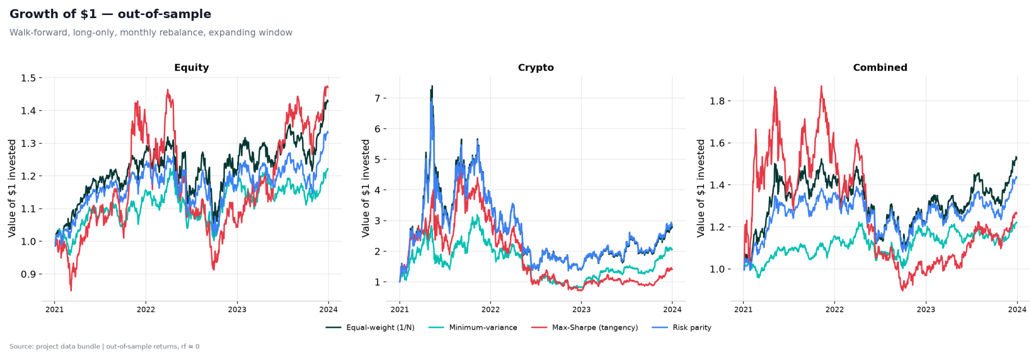

Figure 1. Growth of $1 for all twelve funds. Equity/Combined: 2021-01-04 to 2023-12-29; Crypto: 2021-01-01 to 2023-12-31.

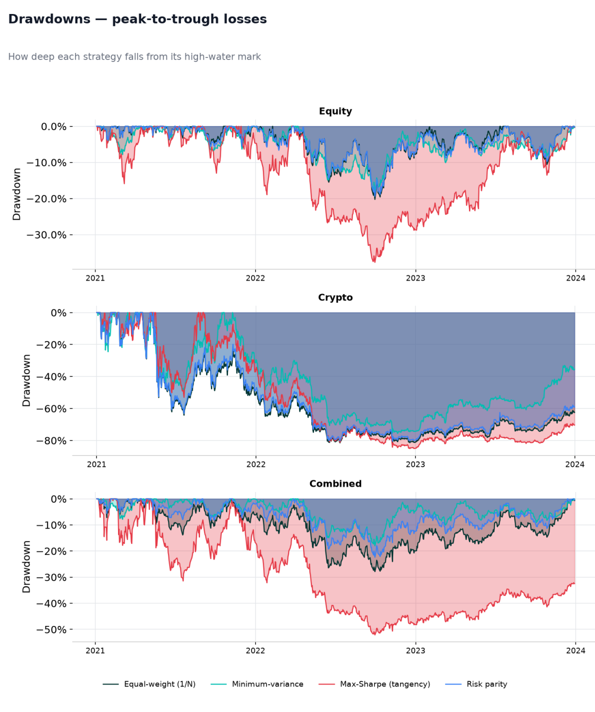

Figure 2. Peak-to-trough drawdowns for all funds. Maximum drawdowns range from −17.99% (Equity and Combined Minimum-variance) to −85.07% (Crypto Maximum-Sharpe).

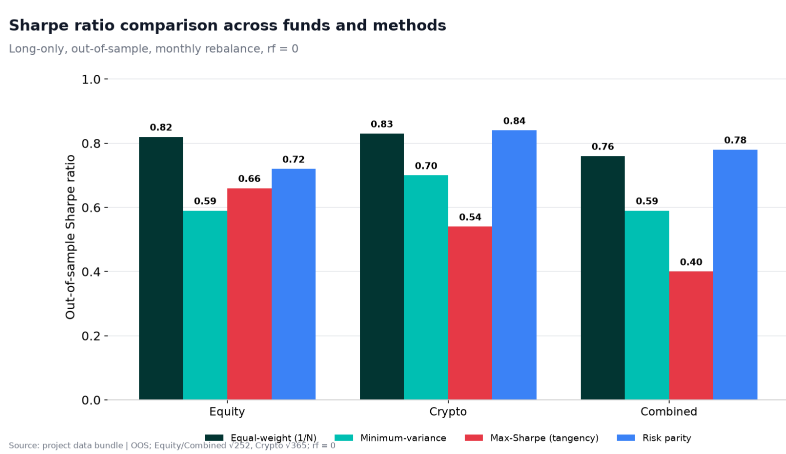

Figure 3. Gross OOS Sharpe ratios using each family's native annualisation convention.

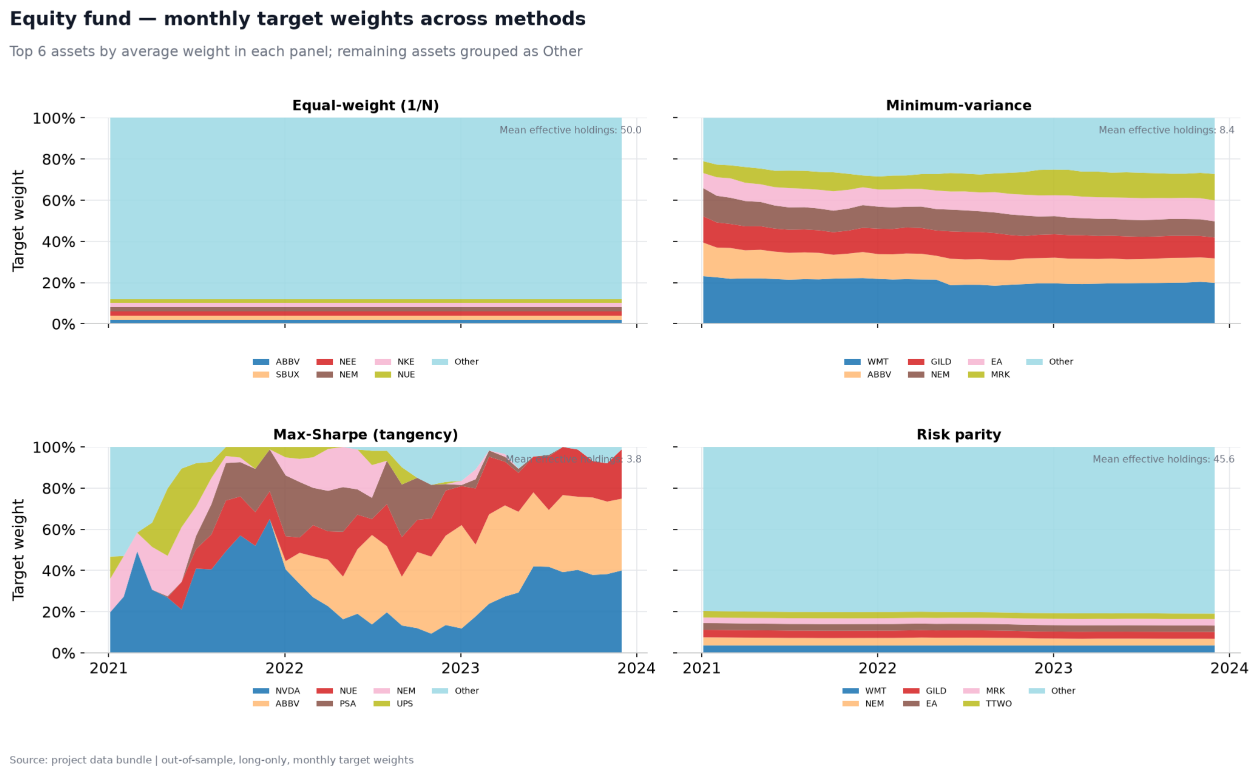

Figure 4. Monthly Equity target weights across the four methods, 2021–2023. Each panel shows the six assets with the highest average target weight for that method; all remaining assets are grouped as Other. Mean effective holdings are $1/\Sigma w_i^2$, averaged across monthly rebalances.

The methods differ markedly in concentration. Equal-weight holds all 50 stocks at 2% at each rebalance, giving an effective holdings count of 50.0. Risk parity is also broad, averaging 45.6 effective holdings and a 16.9% top-five share, because it allocates by covariance-based risk contribution rather than expected return. Minimum-variance is more selective: it averages 8.4 effective holdings, with the five largest positions carrying 66.5% of target weight. Maximum-Sharpe is the most concentrated and variable, averaging only 3.8 effective holdings; its top five average 97.4%, and the largest single position reaches 65.0%. This instability is consistent with its reliance on estimated mean returns and its 27.24% recurring monthly turnover in Table 3. The chart therefore links portfolio construction to implementation: broader methods produce more stable targets, while Maximum-Sharpe's changing concentration helps explain its greater trading-cost sensitivity. These are historical target weights, not current recommendations.

3. The Sentiment Index

3.1 Scoring Model: finVADER

Headlines are scored with VADER (Hutto & Gilbert, 2014), extended with the two finance-specific lexicons distributed with finVADER: SentiBignomics and Henry (Consoli, Barbaglia, & Manzan, 2022; Henry, 2008; Koráb, 2023). This keeps VADER's rules for negation, emphasis, punctuation, and degree modifiers while adding finance vocabulary. Both lexicons are applied to all 146,830 headlines.

3.2 Text Handling Choices

Raw headline text is kept without removing stopwords, changing case, or stripping punctuation because these features affect VADER's rules. For example, negators such as "not", intensifiers such as "very", capitalisation, and punctuation can change a sentence's score. Missing titles are converted to empty strings and score as neutral. Future text and market returns do not influence the score.

3.3 Building the Sector Index

The index is built in three stages. Each headline first receives finVADER compound, positive, negative, and neutral scores. Compound scores are then averaged within ticker-day, preventing heavily covered companies from having a disproportionately large influence on the sector average. Finally, available ticker-day values are equal-weighted within sector-day. Tickers without headlines on a date are excluded rather than imputed as neutral, so the index measures the tone of observed news flow.

For display, the compound score is mapped linearly from [−1, 1] to a Fear & Greed scale from 0 to 100, where 0 denotes extreme fear and 100 extreme greed. This rescaling does not change the underlying analysis; the labels are descriptive, not probabilities or trading signals.

3.4 Look-Ahead Prevention

After sorting by sector and date, the index is shifted by one available observation within each sector. A row dated t therefore contains that sector's score from its previous observed news date, never its current score. At each rebalance, the portfolio uses the latest available lagged value. The first observation for each sector has no lagged value by construction. A unit test builds deliberately unsorted sector observations and checks that every lagged value equals the previous available sector score, not the same-day score.

3.5 Sentiment Patterns

Figure 5 shows the sector Fear & Greed series over 2020–2023, using a 21-day rolling mean so that short-term changes remain readable. In the underlying 9,832 sector-day observations, mean compound sentiment is +0.072; sector means range from +0.058 for Energy to +0.095 for Real Estate. Consumer is the least variable sector (standard deviation 0.064) and Materials the most variable (0.131). These figures describe the supplied headlines and the lexicon used; they may reflect source coverage or calibration and do not show a universal positivity bias in financial news.

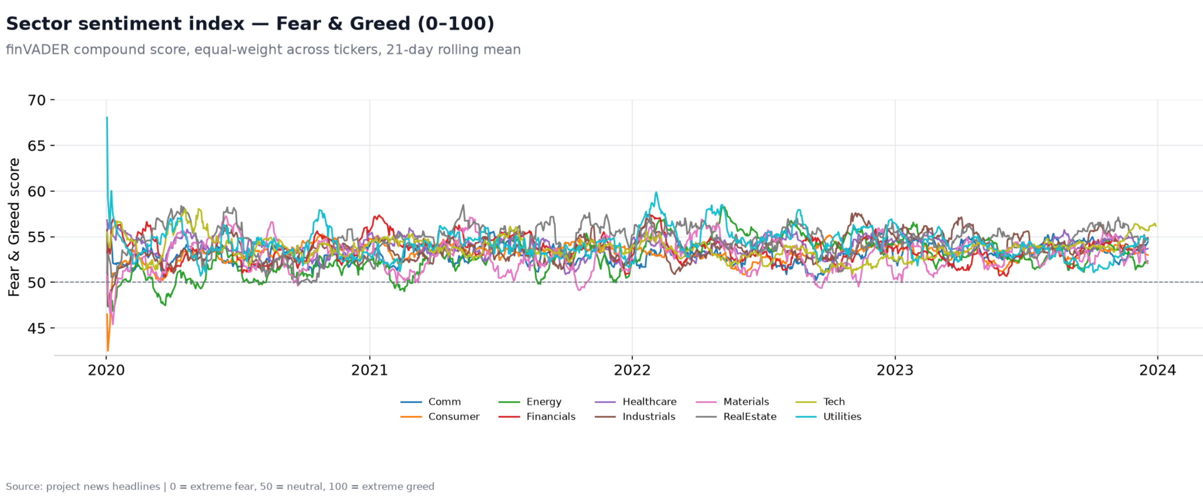

Figure 5. Sector Fear & Greed index, 21-day rolling mean of daily sector scores, 2020–2023. Each daily sector score equal-weights tickers with observed headlines.

4. Extensions and Innovations

The project adds three extensions beyond the required baseline: a finance-adapted VADER lexicon, a lagged sentiment tilt applied to portfolio weights, and a transaction-cost sensitivity based on drifted pre-trade holdings. Each extension is compared with a clear baseline rather than assumed to improve the result.

4.1 Innovation 1: finVADER Extended Finance Lexicon

The two finVADER source lists contain 7,484 entries in total (7,295 SentiBignomics and 189 Henry). After resolving 79 overlapping terms, the project merges 7,405 unique finance entries into VADER; 5,821 are not present in the base VADER lexicon. At headline level, the neutral rate falls from 48.74% under base VADER to 16.79% under finVADER, while mean compound sentiment changes from 0.106 to 0.069. This shows materially different vocabulary coverage, although it does not by itself establish better predictive accuracy.

A small face-validity check used six deliberately clear finance phrases. finVADER assigned positive compound scores to “Revenue beat expectations” (+0.370), “Strong earnings growth” (+0.612), and “Analysts upgrade shares to buy” (+0.315). It assigned negative scores to “Revenue missed expectations” (−0.287), “Weak earnings decline” (−0.612), and “Company files for bankruptcy after severe losses” (−0.436). Base VADER scored “Revenue beat expectations” as neutral (0.000), illustrating one finance-specific coverage difference. The phrase “The firm is profitable” scored +0.026 under finVADER, while “The firm is not profitable” scored −0.017, confirming that VADER’s negation rule remained active after the lexicon merge. This is a face-validity check on selected examples, not an accuracy estimate or evidence that the scores predict returns. Those claims would require a labelled headline sample and a separate forecast test.

4.2 Innovation 2: Sentiment Fusion Tilt

At each monthly rebalance, the most recent available lagged sector sentiment is standardised across the ten sectors. Each Equity Minimum-variance weight is multiplied by (1 + 0.3 × sector z-score), floored at a small positive multiplier, and the portfolio is renormalised to one. The construction preserves long-only full investment and changes only the allocation, allowing a direct comparison with the same base weights and OOS return window.

Table 2 below compares the base Equity Minimum-variance fund against its sentiment-augmented counterpart.

Table 2. Fusion comparison: Equity Minimum-variance base vs. sentiment-tilted (tilt strength = 0.3). Zero transaction costs. OOS period: 2021-01-04 to 2023-12-29.

Fund	Ann. Return	Ann. Vol.	Sharpe	Sortino	Max DD
MinVar (base)	6.92%	12.75%	0.59	0.84	−17.99%
MinVar + sentiment	5.26%	12.77%	0.46	0.66	−18.99%

The sentiment-tilted portfolio underperforms the base across the reported out-of-sample measures: annualised return falls from 6.92% to 5.26%, Sharpe from 0.59 to 0.46, and maximum drawdown worsens from −17.99% to −18.99%. Volatility changes only slightly, from 12.75% to 12.77%, so the weaker outcome comes mainly from lower realised returns. Tetlock (2007) finds that media pessimism is followed by price pressure and then reversal, which indicates that timing matters. This project instead applies a monthly pro-sentiment tilt to sector weights. The rule did not work in this sample. This does not contradict Tetlock (2007), whose study uses different data and signal design.

The experiment is still informative because the rule was specified in advance, lagged to prevent look-ahead, and compared with the same base portfolio over the same dates. It is an original implementation, but the evidence does not show that it improves investment performance.

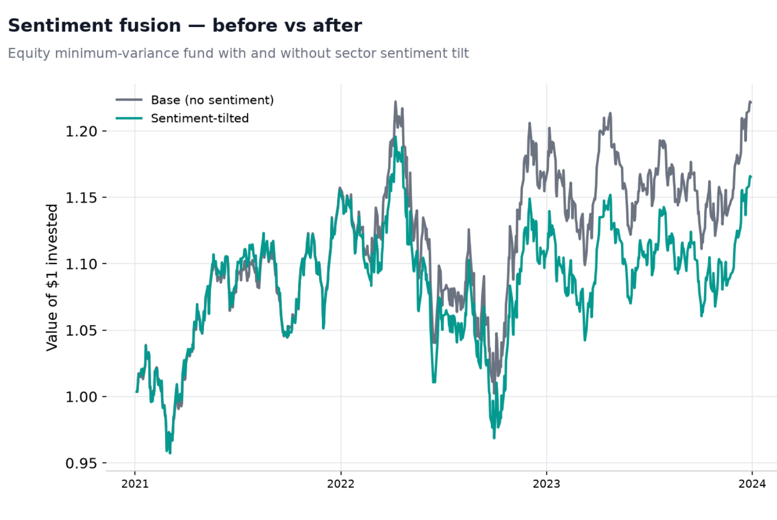

Figure 6. Growth of $1 for Equity Minimum-variance and its sentiment-tilted variant (tilt strength 0.3), 2021-01-04 to 2023-12-29.

4.3 Transaction-Cost Sensitivity

Transaction costs are modelled as a 10-basis-point one-way charge on traded notional. After each monthly target is set, the positions compound without intermediate rebalancing and daily fund returns use the resulting drifted weights. The next rebalance therefore trades from the actual pre-trade holdings to the new target. Turnover is the sum of the absolute trades, the cost is deducted on the rebalance date, and the initial purchase from cash is also charged. This avoids the false zero recurring turnover that target-to-target weights would assign to 1/N. The model remains a proportional-cost sensitivity rather than a complete execution model because bid–ask spreads, nonlinear market impact, taxes, and capacity constraints are outside scope.

Table 3 compares gross and 10-basis-point net performance and reports recurring monthly turnover, where the initial full-investment trade is excluded from the turnover average but included in total cost.

Table 3. Transaction-cost sensitivity using pre-trade drifted weights. Average turnover is recurring monthly traded notional; net results apply a 10 bps one-way proportional charge.

Universe	Method	Avg. turnover	Gross ann. ret.	Net ann. ret.	Gross Sharpe	Net Sharpe
Equity	Equal-weight	5.45%	12.64%	12.53%	0.82	0.81
Equity	Min-variance	6.12%	6.92%	6.81%	0.59	0.58
Equity	Max-Sharpe	27.24%	13.77%	13.37%	0.66	0.65
Equity	Risk parity	5.16%	10.07%	9.97%	0.72	0.71
Crypto	Equal-weight	11.90%	40.50%	40.27%	0.83	0.83
Crypto	Min-variance	4.73%	27.16%	27.05%	0.70	0.69
Crypto	Max-Sharpe	13.60%	12.08%	11.86%	0.54	0.54
Crypto	Risk parity	12.00%	41.79%	41.55%	0.84	0.84
Combined	Equal-weight	7.71%	15.14%	14.99%	0.76	0.76
Combined	Min-variance	6.12%	6.92%	6.81%	0.59	0.58
Combined	Max-Sharpe	30.85%	8.13%	7.71%	0.40	0.39
Combined	Risk parity	6.48%	12.79%	12.67%	0.78	0.77

Maximum-Sharpe is the most sensitive to transaction costs because it trades most aggressively: recurring monthly turnover averages 27.24% in Equity and 30.85% in Combined, reducing annualised return by about 40 and 42 basis points respectively. Equal-weight and Risk parity still trade after their weights drift, but their annualised return drag is about 10–24 basis points. At the assumed 10 bps charge, costs do not change the broad strategy rankings. This conclusion depends on the simplified linear cost assumption.

5. The App and the Investor Journey

5.1 Target User and Product Purpose

Quantise is designed for a financially literate retail investor or small family office seeking transparent, systematic exposure to equities, cryptoassets, or a combination of both. The intended user understands measures such as annualised return, volatility, Sharpe ratio, and maximum drawdown, but may not have the data infrastructure or technical expertise required to construct and evaluate optimised portfolios independently.

The application therefore prioritises interpretability. For each fund family, it exposes the investment objective, asset universe, optimisation method, latest available historical portfolio weights, out-of-sample performance, and risk measures instead of presenting the model as a black box. Quantise remains an academic research prototype rather than an investment product: its backtests, sentiment indicators, and illustrative allocations are not forecasts, recommendations, or personal financial advice.

5.2 How the App Is Organised

The Streamlit application has four pages: Home, Funds, Sentiment, and Data Explorer. Sidebar navigation, the Quantise wordmark, and consistent page styling help users recognise where they are. The Home page introduces the research, summarises the dataset, and links directly to the main analysis pages and tools, so it acts as the starting point for the workflow rather than a static summary.

The Funds page is the main portfolio workspace. Users choose the Equity, Crypto, or Combined family and the strategies they want to examine. The page then shows growth of $1, drawdowns, risk-adjusted performance, transaction-cost results, and the latest historical rebalance weights. It also includes Fund Comparison, an illustrative allocation builder, the Correlation Explorer, and the Risk Scenario Explorer.

The Sentiment page focuses on the headline analysis. A sector selector lets users compare the 10 Equity sectors through the latest Fear & Greed score, its recent change and average, a time series, and a monthly heatmap. A headline browser appears when the local headline file is available. The Data Explorer provides sortable and filterable records for fund returns, portfolio weights, performance metrics, and sector sentiment, with CSV downloads. Individual headline records are available locally but omitted from the lightweight cloud deployment because of their size.

5.3 Interactive Investor Journey

The application follows four stages: Discover, Evaluate, Build, and Compare. In Discover, the user chooses a fund family and reviews its purpose, assets, risk profile, and methods. In Evaluate, the user examines walk-forward returns, volatility, drawdowns, risk-adjusted measures, transaction costs, and historical weights. Short guide panels explain measures such as Fear & Greed and correlation beside the relevant charts.

In Build, the user combines chosen fund families and strategies into an illustrative allocation and sees its historical growth and risk. In Compare, two alternatives can be viewed side by side, their correlations examined over a selected period, and risk or sentiment scenarios explored. These tools help users explore trade-offs; they do not choose a portfolio for them.

The journey bar on the Funds page is interactive: selecting a stage moves the user to the corresponding part of the workflow. Home-page buttons and feature cards also link to the relevant pages or tools. These links connect the introductory explanations with the detailed analysis without forcing the user to search through the full page.

5.4 Why the Interactive Features Matter

Each interactive feature is tied to a practical question. Fund and strategy selectors define what the user wants to study, while the linked growth, drawdown, weight, and risk views show different aspects of the same result. Fund Comparison places two choices in one view and states their observation periods and annualisation conventions, making differences between Equity-calendar and Crypto-calendar results visible.

The Correlation Explorer lets the user change the analysis period and check whether relationships differ across subperiods instead of relying only on one full-sample matrix. The allocation builder converts chosen fund and strategy weights into historical portfolio results. The Risk Scenario Explorer uses three short risk-preference questions to produce an illustrative historical allocation; it is an educational scenario, not a suitability test or personal recommendation.

The sector selector, sentiment time series, monthly heatmap, and optional headline filters move from a summary score to the underlying news evidence. The What-If Tilt Slider blends historical daily returns between the base Equity Minimum-variance strategy and the implemented 0.3 sentiment tilt. Intermediate settings are sensitivity illustrations, not separately optimised backtests, and the slider does not extrapolate beyond the two tested endpoints.

The Data Explorer lets users inspect, sort, filter, and download the saved data behind the charts. These actions do not rerun sentiment scoring or portfolio optimisation, which keeps the deployed app responsive and separates result generation from presentation.

5.5 Deployment and Disclosures

The application is deployed through Streamlit Community Cloud from the repository's root-level streamlit_app.py file. It reads saved outputs from the results directories instead of rerunning sentiment scoring or portfolio optimisation. This keeps the public app lightweight, while the local pipeline retains the data preparation, walk-forward estimation, optimisation checks, and transaction-cost calculations needed to reproduce the results.

The interface identifies Quantise as a FINS3645 academic research prototype based on historical walk-forward out-of-sample backtests. It explains that sentiment scores measure the language of news headlines rather than market prices, scenarios are illustrative, past performance does not guarantee future outcomes, and actual trading costs may differ from the model. These notes appear beside the relevant charts and in the sidebar.

6. Critical Reflection and Recommendations

6.1 What Worked

A key technical improvement is the more reliable research pipeline. Equity and Crypto retain their own calendars, each portfolio decision uses only earlier data, and failed or invalid optimiser solutions cannot pass silently. Transaction costs are based on holdings after they drift with returns, not simply on the difference between consecutive target weights. These choices address the main technical risks found during development: calendar compression, look-ahead, silent solver failure, and understated turnover.

The results also show why simple benchmarks matter. Equal-weight has the highest Equity Sharpe ratio, while Risk parity has the highest Sharpe ratio in Crypto and Combined. Combined Minimum-variance gives Crypto zero weight within numerical precision, so its results are identical to Equity Minimum-variance at the reported precision. These sample-specific findings show that a more complex method or a wider asset set does not guarantee better realised diversification. DeMiguel, Garlappi, and Uppal (2009) likewise find that estimation error can offset the theoretical benefits of optimised portfolios relative to 1/N out of sample.

The finance-adapted lexicon is a useful extension because it recognises many terms missed by base VADER and produces a sector index with visible variation. The evidence supports a claim of broader finance vocabulary coverage, not higher predictive accuracy. The project has neither manually labelled headline sentiment nor a formal prediction test that could establish that finVADER is more accurate than base VADER.

6.2 Limitations and What Did Not Work

The sentiment-fusion rule underperforms its directly comparable Equity Minimum-variance baseline on every reported out-of-sample measure. Because the signal is lagged and the result is reported despite being unfavourable, the comparison is still useful. It shows that this particular monthly sector-level tilt did not add value over 2021–2023; it does not show that all uses of financial-news sentiment will fail.

The experiment cannot identify the reason for the underperformance. Averaging by sector may remove firm-specific information, monthly rebalancing may be too slow, or the score may describe current news without predicting later returns. Tetlock (2007) links media pessimism to subsequent price pressure, reversal, and trading volume, but his study uses a different news source, signal, and horizon. It therefore motivates further testing rather than validating this allocation rule.

The three-year out-of-sample period is another limitation. It is only one historical path, and the report does not estimate uncertainty around differences in returns or Sharpe ratios. Rankings may change with the sample window, estimation length, rebalance frequency, or market regime. The fixed 10-basis-point cost is also a sensitivity test rather than an execution forecast because it excludes bid–ask spreads, changing liquidity, nonlinear market impact, taxes, and capacity limits.

The app has been checked for correct operation but not tested formally with users, so the report cannot claim that it improves understanding. The full headline file is also available only in the local version; the public app does not expose every research record.

6.3 Three Concrete Recommendations

Recommendation 1: Check whether the strategy rankings are robust. Extend the out-of-sample analysis over a longer history and vary the estimation window, rebalance frequency, and market subperiod. Compare every optimised method with 1/N and simple asset-class benchmarks, and report uncertainty measures that allow for serial dependence, such as block-bootstrap intervals. This would show whether the rankings persist beyond the 2021–2023 sample and directly addresses the estimation-error concern in DeMiguel et al. (2009).

Recommendation 2: Test whether sentiment predicts returns before designing another tilt. Examine whether lagged sentiment explains later ticker or sector returns at horizons chosen in advance. Compare base VADER with finVADER, ticker-level with sector-level aggregation, and different lag or decay windows. Momentum and contrarian rules should be treated as competing hypotheses rather than selected after portfolio results are known. This separates broader lexicon coverage from return prediction and portfolio construction.

Recommendation 3: Use more realistic trading costs and test the interface with users. Add observed bid–ask spreads and a volume-sensitive impact component to the proportional cost model. Almgren and Chriss (2001) distinguish permanent and temporary market impact and formalise the trade-off between execution cost and price risk. For the app, provide a deployable headline sample or external data store, then test whether users can compare risk and return, interpret sentiment, find supporting data, and recognise that scenario outputs are illustrations rather than recommendations.

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